Stepper motor

DO | PICO | OUTPUT | MOVEMENT

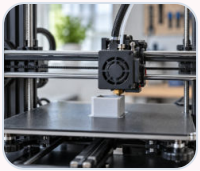
What does it do?

A stepper motor turns in small, repeatable steps instead of spinning freely.

Energising its coils in sequence moves the shaft one step at a time and holds its position.

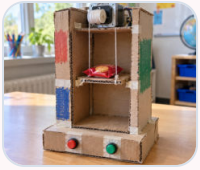
The ULN2003 board supplies the coil current for the common 28BYJ-48 motor shown here.

Uses



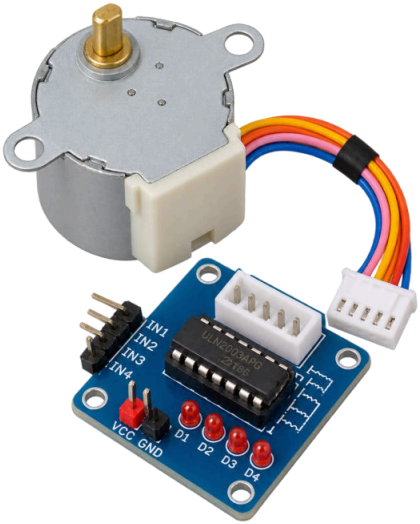
3D printer positioning

DETECT Read next print position → **DECIDE** How many steps are needed? → **DO** Move the print head



Snack lift

DETECT Snack request arrives → **DECIDE** Which floor needs it? → **DO** Step lift into place



Output shaft Moves in measured steps

Five-wire plug Connects to the driver

IN1 to IN4 Coil sequence

5V and GND Driver power

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import Stepper

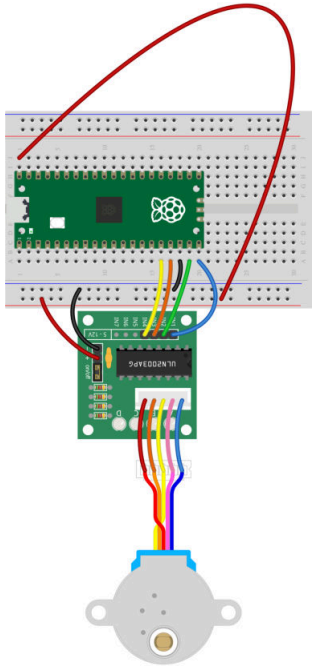
motor = Stepper((15, 14, 13, 12))

motor.step(100)
```

Call `motor.step(100)` again to keep moving. Add `direction="ccw"` to turn the other way.

Wiring

PICO



IN1-IN4 → GP **15, 14, 13, 12** (pins 20, 19, 17, 16); driver → **VBUS** (pin 40, bottom + rail) and **GND** (pin 18)